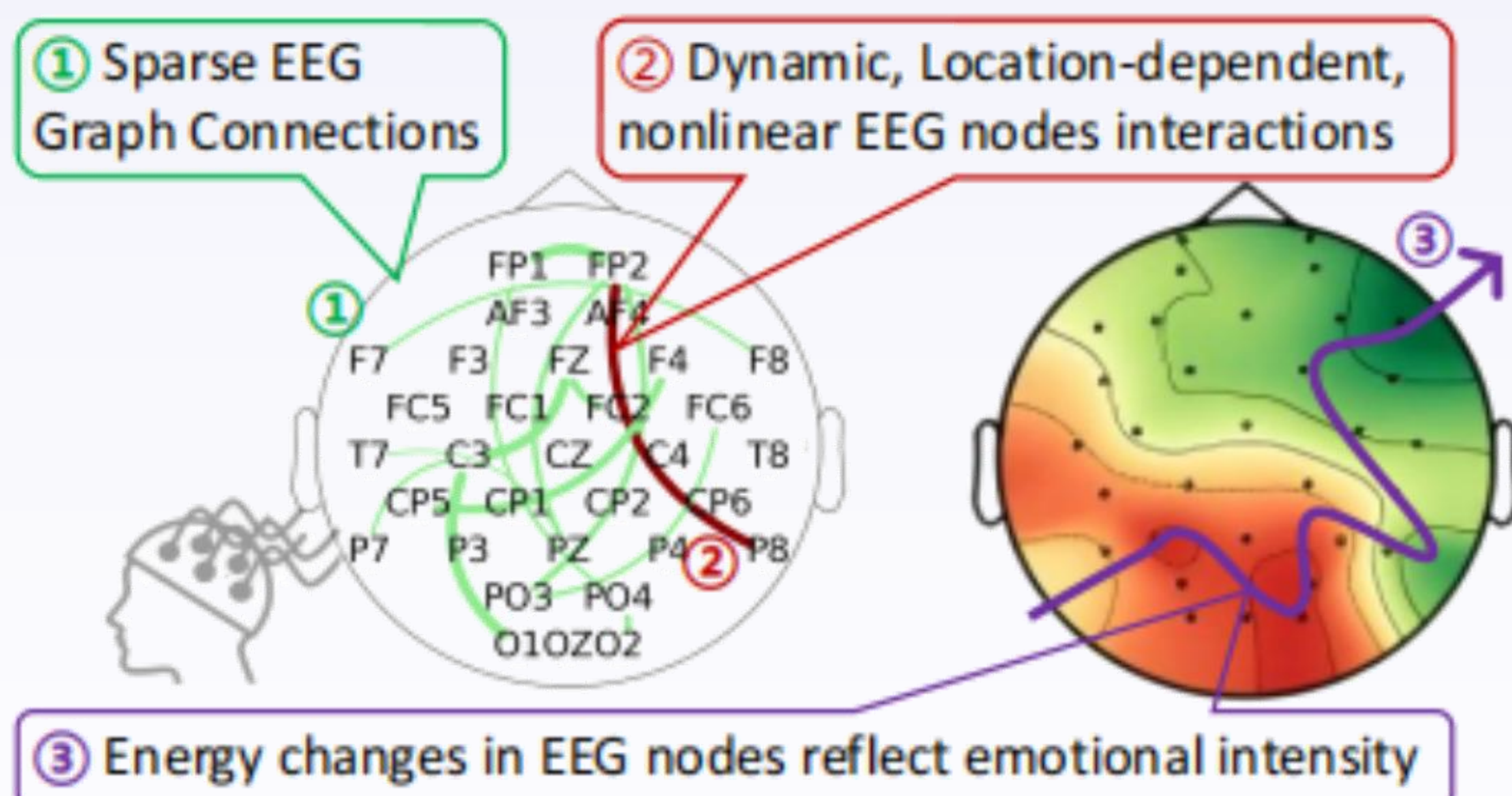


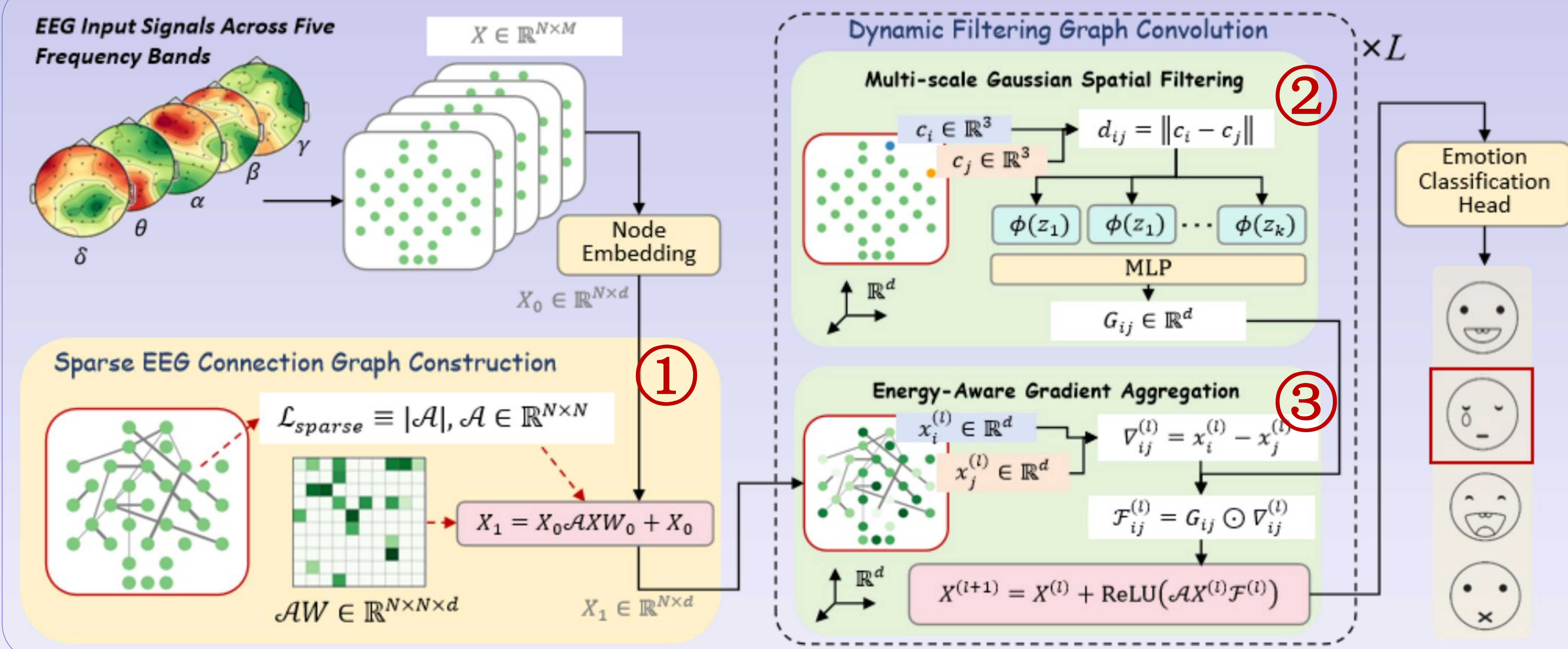
Overview

- Background:** Emotional experiences are products of psychological events formed by coordinated interactions between cortical and subcortical neural networks.
- Challenges:** How these interactions dynamically adapt to spatial and spectral changes in neural signals, and how these dynamics contribute to emotional experiences, remain inadequately understood.
- Our Solutions:** we propose a novel framework, Spatial-Energy-Aware Dynamic Filtering with Sparse Graph Convolutions (SEASGC). Our approach rethinks EEG graph modeling from three key aspects:
 - Incorporating sparse EEG graph connections to better capture the essential functional relationships between brain regions**
 - Modeling dynamic, location-dependent, nonlinear interactions between EEG nodes**
 - Leveraging energy changes in EEG nodes as critical indicators of emotional intensity**



- Goal:** By explicitly integrating these principles, our method aims to provide a more comprehensive representation of EEG signals for emotion recognition.

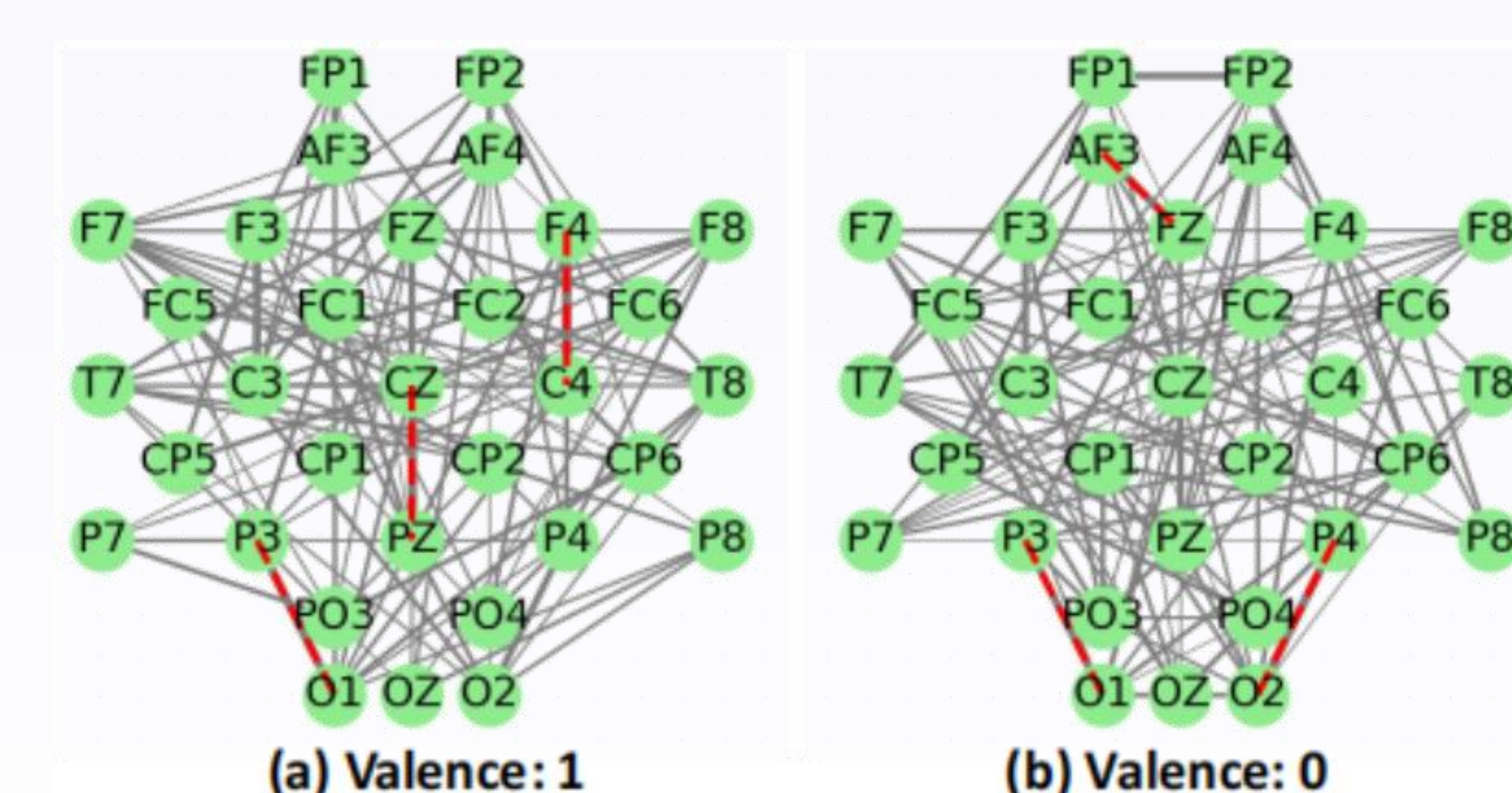
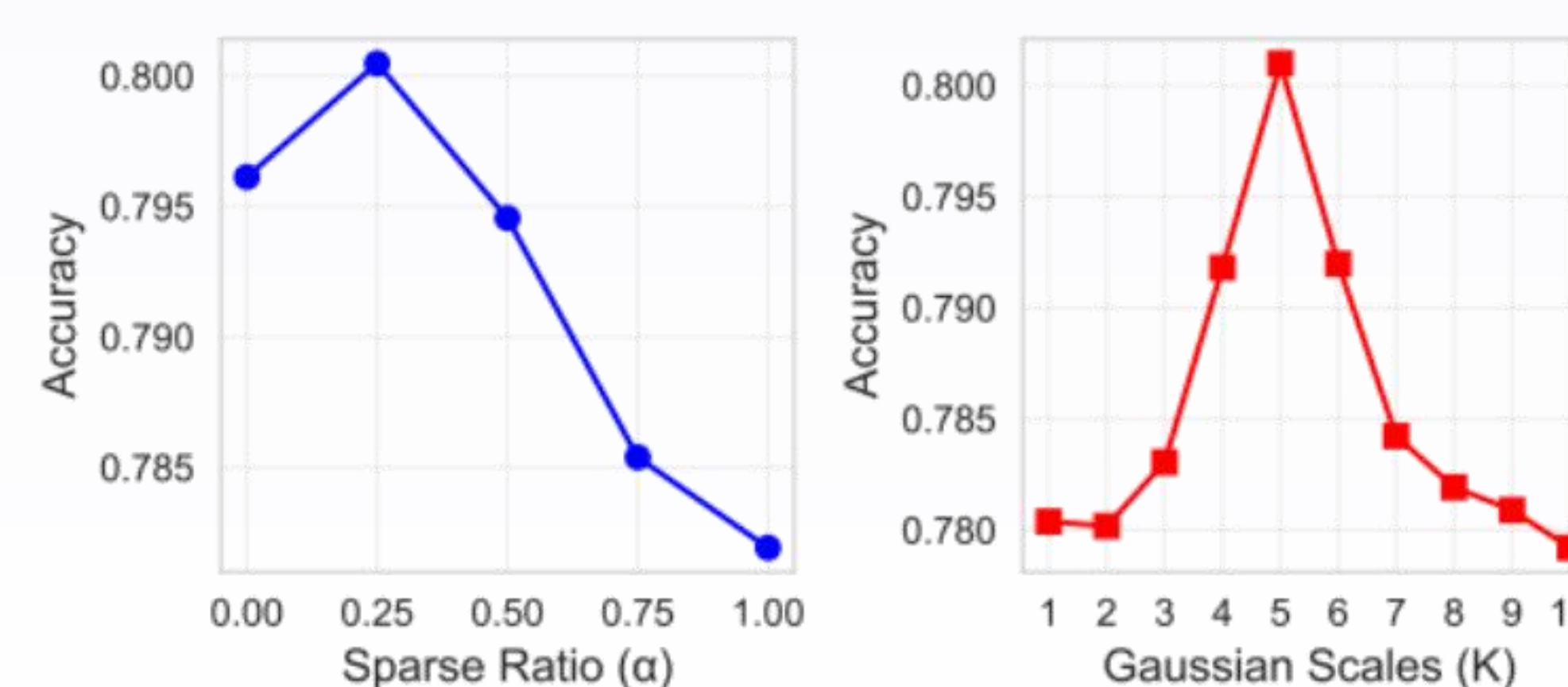
Methods



Results

Methods	DEAP				DREAMER			
	Arousal		Valence		Arousal		Valence	
DGCNN (Song et al., 2018)	64.89 ± 8.73	66.17 ± 10.16	68.14 ± 6.72	72.89 ± 6.44	68.56 ± 8.51	67.85 ± 9.55	68.00 ± 6.51	63.95 ± 7.11
MM-ResLSTM (Ma, Tang, Zheng, & Lu, 2019)	70.84 ± 9.48	68.07 ± 10.81	70.89 ± 7.64	64.21 ± 7.12	68.43 ± 7.25	63.77 ± 7.31	71.34 ± 6.72	70.63 ± 5.39
ACRNN (Tao et al., 2020)	69.71 ± 8.41	65.48 ± 8.04	68.82 ± 6.23	69.33 ± 7.09	70.24 ± 9.26	65.78 ± 9.54	65.23 ± 5.11	60.27 ± 6.99
MTGNN (Wu et al., 2020)	70.98 ± 7.37	68.45 ± 9.98	69.14 ± 6.47	68.97 ± 6.19	71.24 ± 7.26	69.87 ± 9.53	67.07 ± 5.93	68.07 ± 8.31
SST-EmotionNet (Jia et al., 2020)	73.49 ± 6.61	76.84 ± 10.72	70.94 ± 6.21	69.87 ± 6.99	71.54 ± 8.43	69.83 ± 8.71	70.39 ± 7.03	68.10 ± 5.27
HetEmotionNet (Jia et al., 2021)	74.36 ± 7.85	75.71 ± 10.82	72.08 ± 5.66	71.15 ± 6.1	70.58 ± 7.63	69.24 ± 4.92	70.03 ± 6.64	67.49 ± 6.38
ASTG-LSTM (Li, Zheng, Zong, Chang, & Lu, 2021)	72.55 ± 6.83	73.18 ± 11.01	71.96 ± 6.48	70.8 ± 7.26	70.23 ± 8.43	69.14 ± 5.34	68.76 ± 6.76	68.94 ± 5.49
RGNN (Zhong et al., 2022)	60.55 ± 5.31	62.49 ± 7.14	61.37 ± 6.93	60.75 ± 6.16	62.48 ± 5.03	61.07 ± 6.47	61.98 ± 5.87	61.11 ± 5.37
TSception (Ding et al., 2022)	74.38 ± 8.18	76.21 ± 9.84	68.94 ± 7.25	72.04 ± 6.94	69.77 ± 6.56	69.27 ± 9.43	68.16 ± 7.12	65.06 ± 6.71
ST-GCLSTM (Feng, Cheng, Zhao, Deng, & Zhang, 2022)	76.68 ± 7.37	71.34 ± 9.85	71.81 ± 6.72	70.36 ± 9.14	69.29 ± 6.21	69.43 ± 5.93	70.49 ± 5.98	70.23 ± 5.84
LGGNet (Ding et al., 2024)	72.31 ± 8.38	73.28 ± 8.78	68.53 ± 7.79	69.15 ± 7.59	70.49 ± 8.84	68.94 ± 8.47	68.69 ± 6.08	66.12 ± 8.14
VSGT (Liu et al., 2024)	78.95 ± 6.89	78.65 ± 8.24	74.15 ± 6.41	74.05 ± 6.43	75.19 ± 6.87	72.98 ± 7.54	73.41 ± 6.36	72.73 ± 5.71
SEASGC (Ours)	80.03 ± 5.94	79.96 ± 7.35	77.12 ± 5.63	77.01 ± 5.77	76.97 ± 5.91	73.89 ± 6.42	75.22 ± 5.37	74.56 ± 5.12

Model	DEAP (Arousal)		DEAP (Valence)	
	Accuracy	F1-score	Accuracy	F1-score
Full SEASGC	80.03	79.96	77.12	77.01
w/o SGC	77.15	76.88	74.65	74.21
w/o MSGF	77.13	77.09	75.15	75.04
w/o EAGA	78.04	77.77	75.88	75.63
w/o MSGF & EAGA	75.98	75.34	73.42	73.01



Brain region connections for Valence Label 1 (positive emotions) and Valence Label 0 (negative emotions)